

# Effects of Isocaloric Resistance, Aerobic, and Concurrent Exercise on Excess Postexercise Oxygen Consumption in Older Adults

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## Abstract

Pilon, R, Matos-Santos, L, Matlez, MP, Rodrigues, G, Amorim, F, Lattari, E, Farinatti, P, and Monteiro, W. Effects of isocaloric resistance, aerobic, and concurrent exercise on excess postexercise oxygen consumption in older adults. *J Strength Cond Res* 38(4): 755–761, 2024—Excess postexercise oxygen consumption (EPOC) is a major determinant of exercise-related caloric expenditure and metabolic adaptations. Exercise modality may influence the EPOC, but this issue has not been investigated in older adults. This study compared the EPOC after isocaloric bouts of continuous aerobic exercise (AE), resistance exercise (RE), and concurrent exercise (CE) in older individuals. Ten subjects (5 men;  $73 \pm 6$  years) had their cardiorespiratory data assessed during AE, RE, and CE and along 30-minute postexercise recovery. Total energy expenditure (EE) during exercise was similar ( $p > 0.05$ ) in AE ( $126.0 \pm 30.7$  kcal), RE ( $123.9 \pm 30.6$  kcal), and CE ( $130.8 \pm 32.6$  kcal), with different times to achieve the targeted EE (RE:  $61.4 \pm 1.9$  minutes  $>$  CE:  $43.3 \pm 5.6$  minutes  $>$  AE:  $26.6 \pm 5.7$  minutes;  $p < 0.001$ ). Consistently, the relative intensity during exercise was superior ( $p < 0.05$ ) in AE ( $74 \pm 15\%$  oxygen uptake reserve [ $\dot{V}O_{2R}$ ]) vs. CE ( $43 \pm 13\% \dot{V}O_{2R}$ ) vs. RE ( $24 \pm 9\% \dot{V}O_{2R}$ ). Despite the isocaloric conditions, average EPOC and EE were approximately 45% greater ( $p < 0.001$ ) in AE ( $8.0 \pm 2.3$  L;  $40.1 \pm 11.7$  kcal) vs. RE ( $5.6 \pm 1.2$  L;  $28.1 \pm 5.8$  kcal) and CE ( $5.4 \pm 2.3$  L;  $26.9 \pm 11.5$  kcal). In conclusion, the EPOC was greater after isocaloric AE vs. RE and CE performed by older adults. Exercise intensity seemed to be a more important determinant of EPOC than volume reflected by EE during exercise bouts. Moderate-intensity continuous AE was more time-efficient than RE and CE to achieve a target EE. In older individuals, AE should be preferred over RE or CE when the purpose is to increase the daily caloric expenditure.

**Key Words:** oxygen consumption, cardiorespiratory fitness, muscle strength, energy expenditure, biological aging

## Introduction

Older adults often exhibit decreases in muscle mass and strength, which compromises their functional independence (36). These changes may reduce the energy expenditure (EE) at rest (37,38) and during physical activity (19), predisposing to lowered resting metabolic rate (RMR) (37), increased fat mass (41), and higher cardiovascular risk (33). Physical training counteracts those effects, not only because of its benefits on muscle mass (15) but also because of an increase in daily EE (19,42).

Transient elevations in metabolic rate underpin the exercise-related increase in EE, usually referred to as excess postexercise oxygen consumption (EPOC) (21,26). The determinants of EPOC depend on exercise modality. In resistance training, the oxygen uptake ( $\dot{V}O_2$ ) remains elevated during recovery because of factors as diverse as the number of exercises, recruited muscle, intervals between sets and exercises, and workload (9,12,38). However, the EPOC following aerobic and resistance exercise seems to result from a combination of exercise intensity and duration (24,31).

A few studies compared the EPOC elicited by resistance and aerobic exercise. In general, the magnitude of  $\dot{V}O_2$  assessed from 30 minutes to 5 hours of recovery has been reported to be superior

after resistance than aerobic exercise, even when the bouts are matched for the overall EE (5,17,38). A single trial compared isocaloric resistance vs. aerobic exercise sessions performed with different intensities (moderate continuous and high intermittent) (18)—the EPOC assessed during 12 hours was greater after resistance vs. moderate continuous but similar to high-intensity intermittent exercise. Noteworthy, those studies investigated only young adults. We could not find trials comparing the EPOC after resistance and aerobic exercise in older individuals. This would be nonetheless useful given the fact that age-related differences in body composition (30) or the ability to perform prolonged and intense efforts (30,35) may affect the metabolic responses after exercise (18). Moreover, previous experiments adopted strategies to “burst” the increase of heart rate and  $\dot{V}O_2$  during resistance exercise. Systems such as supersets or high-intensity circuits with very short recovery intervals have been applied, which is not usual in conventional exercise training designed for older people (9,18).

Another interesting issue refers to concurrent training, which has been endorsed for a wide range of populations, including older adults (15,22). This modality is advantageous because it aggregates the benefits of both resistance and aerobic training. We found a single trial comparing the  $\dot{V}O_2$  and total EE during 3 different circuit weight-training protocols of equivalent duration, composed of free weight exercises, machine exercises, and a combination of free weight and aerobic exercises (3). The EE was 20–30% higher in combined vs. free weight and machine

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protocols, suggesting that a combination of resistance and aerobic exercises induced greater overall training volume than exclusive resistance training. However, we did not locate previous studies comparing the EE elicited by concurrent vs. isolated aerobic exercise or evaluating the EPOC after resistance, aerobic, or concurrent exercise (CE).

In short, given the lowered RMR typically observed in older adults and considering that physical exercise may increase the overall EE, research on the EPOC in this population is warranted. Comparative studies after acute aerobic, resistance, and CE are limited and included only young individuals. Moreover, data from resistance exercise were often obtained in artificial research sets, in which the intensity and volume of exercise protocols were unusually high. The present study addressed those gaps by comparing the EPOC after isocaloric bouts of aerobic, resistance, and CE complying with the current recommendations for older adults. We hypothesized that concurrent training would induce greater EPOC in comparison with resistance or aerobic training performed in isolation, even in protocols matched for the overall EE.

## Methods

### Experimental Approach to the Problem

This study included 5 visits interspersed with at least 48 hours to avoid the confounding effects of acute exercise on the metabolic rate. The experimental sessions occurred always in the morning (7–11 AM) to negate circadian variation in EPOC: (a) resistance exercise (RE); (b) aerobic exercise (AE); and (c) CE (CE = RE + AE). On the first visit, subjects signed informed consent and underwent clinical examination and anthropometric measurements. After this, they remained sit at rest for 20 minutes to mimic the duration of exercise bouts. Subsequently, resting  $\dot{V}O_2$  was assessed for 30 minutes at the supine position after strict recommendations (6). Data from the last 5-minute window were extracted to determine the RMR (7). Finally, a cycle ergometer maximal cardiopulmonary exercise testing (CPET) was performed to determine the peak  $\dot{V}O_2$  ( $\dot{V}O_{2peak}$ ) and heart rate ( $HR_{peak}$ ). On the second visit, loads corresponding to 10 repetition maximum (RM) were determined for the exercises included in RE (2). Then, subjects performed a 10-minute adaptation with the silicone facemask (Hans Rudolph, Kansas, MO) used to assess the  $\dot{V}O_2$  during RE, AE, and CE.

On the next visits, isocaloric exercise sessions (RE, AE, and CE) interspersed with 48-hour intervals were performed within 7–10 days. The RE was always placed first and included 8 exercises for the upper and lower body with loads corresponding to 80% of 10 RM (15). The total EE individually assessed during RE was used as the target for AE and CE sessions, whose order was randomized. This strategy was adopted because of difficulties in pre-defining the EE to be achieved during RE (24). The AE consisted of a continuous cycle exercise performed at 60–65% of  $\dot{V}O_2$  reserve ( $\dot{V}O_{2R}$ ) until reaching the targeted EE. The CE session included exercises performed in AE and RE. To assure the isocaloric condition, the RE and AE components should elicit half of the target EE applied when those modalities were performed in isolation.

Before the exercise sessions, the  $\dot{V}O_2$  was assessed at rest for 10 minutes in a supine position (pre-exercise). A variation of 5% from the resting  $\dot{V}O_2$  assessed on the first visit was accepted. In the case of greater variation, subjects should return another day but this situation did not occur. After the exercise sessions,

subjects were placed supine and the  $\dot{V}O_2$  was measured for an additional 30 minutes. The EPOC and EE were calculated after RE, AE, and CE, as detailed elsewhere (43). All assessments occurred in similar conditions of temperature (18–22° C), air humidity (50–75%), and barometric pressure (750–765 mm Hg). Figure 1 summarizes the study design.

### Subjects

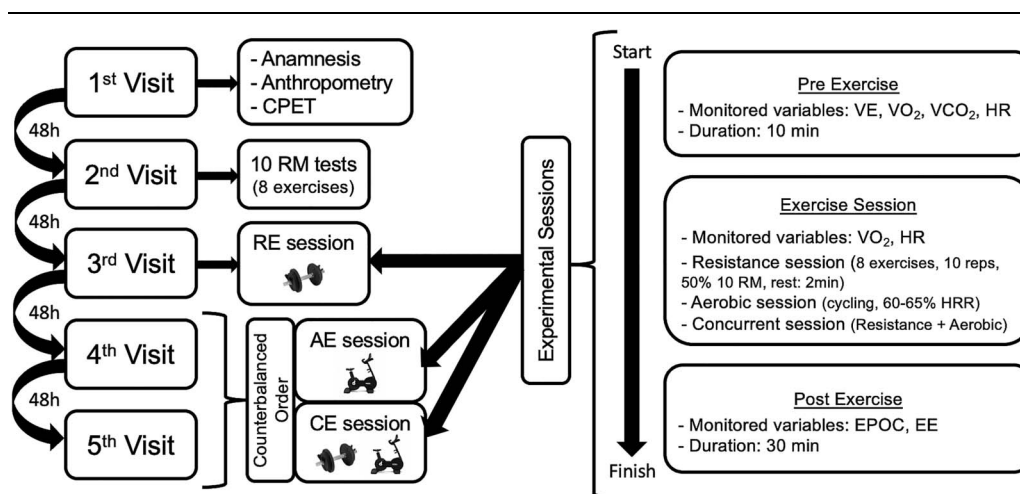
Subjects were recruited through advertisements on social media and lectures at an exercise program for older adults developed at the University facilities. To be eligible, they should be at least 60 years and have medical clearance for physical exercise. Moreover, they should have at least 6 months of previous experience with RE and AE and not use medications or supplements influencing the metabolism (e.g., oral steroids, anxiolytics, herbal supplements, and hormonal or stimulant drugs). Individuals with any musculoskeletal, cardiovascular, or neurological impairments limiting the ability to complete the exercise protocols were excluded from the study.

Initially, 31 older adults volunteered. Nine of them were under cardiovascular medication that could interfere with cardiorespiratory responses; 6 exhibited joint problems precluding the completion of exercises; 2 did not have the minimum required experience with RE or AE. Among the eligible subjects, 2 failed to attain the criteria for maximum CPET; 1 did not complete all exercise protocols; 1 dropped out because of personal problems. The final sample consisted of 10 individuals aged 63 to 82 years (5 women,  $72.9 \pm 6.0$  years,  $70.5 \pm 9.9$  kg,  $161.1 \pm 8.0$  cm). All subjects signed informed consent and the experiment gained approval from the Salgado de Oliveira University institutional ethics committee (CAAE: 71963417.7.0000.5289).

**10 Repetition Maximum Assessment.** The 10 RM tests and retests were performed on the same day, interspersed by 1 hour. Loads corresponding to 10 RM were determined for 8 exercises, in the following order (2): leg press, seated row, leg extension, seated bench press, leg curl, seated machine shoulder press, seated machine hip adductor, and standing cable biceps curl (TechnoGym, Gambettola, Italy). The same experienced professional conducted the assessments (R.P.). Each test was preceded by a warm-up consisting of 1 set of 15 repetitions with approximately 30–40% of body mass. Three attempts interspersed with 5-minute intervals were allowed for each exercise, and the highest value was recorded as result.

The 10 RM load was considered correct when subjects were not able to perform the 11th repetition. In this case, subjects performed further attempts with loads incremented by 5–10%. Despite the cadence not being controlled, subjects were instructed to perform the concentric phase as fast as possible and the eccentric phase in a controlled manner. Intervals between tests of consecutive exercises were interspersed with 10-minute intervals. The 10 RM retest for determination of data reliability was conducted following the same protocol. However, warm-up loads corresponded to 30–40% of 10 RM loads achieved on the first day. Verbal encouragement was given to maintain high motivation during tests and retests.

**Cardiopulmonary Exercise Testing.** The CPETs were performed on an electronically braked cycle ergometer (Inbramed, Porto Alegre, Brazil) using a ramp protocol to determine the  $\dot{V}O_{2peak}$ . Final work output and work rate increments were calculated



**Figure 1.** Overview of study design. CPET = cardiopulmonary exercise testing; 10 RM = 10 repetition maximum; RE = resistance exercise; AE = aerobic exercise; CE = concurrent exercise; HR = heart rate; HRR = heart rate reserve;  $\dot{V}O_2$  = oxygen consumption;  $\dot{V}CO_2$  = carbon dioxide production; EE = Energy Expenditure; EPOC = excess postexercise oxygen consumption.

using the ACSM's metabolic equations for cycling and walking (2) based on a target time to exhaustion of 10 minutes. The work rate for the 3-minute warm-up period was 0 W and the initial work rate was 30 W. Cycling cadence was maintained at 55–65 revs  $\text{min}^{-1}$ . Tests were considered maximal if at least 3 of the following criteria were satisfied (20): (a) 90% of predicted  $\text{HR}_{\text{max}}$  or  $\Delta\text{HR}$  between 2 consecutive work rates  $\leq 4 \text{ b} \cdot \text{min}^{-1}$ ; (b)  $\Delta\dot{V}O_2$  between 2 consecutive work rates  $< 2.1 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ ; (c) respiratory exchange ratio (RER)  $> 1.10$ ; (d) voluntary exhaustion as reflected by a score of 9 on the Borg CR10 scale.

The heart rate was continuously measured using a cardio tachometer (RS800cx; Polar, Kempele, Finland) and beat-by-beat data were 30-second stationary time-averaged. Gas exchange data were acquired every 3 complete respiratory cycles ( $\sim 10$  seconds) via metabolic cart ( $\dot{V}O_{2000}$ ; Medical Graphics, St. Paul, MN). A facemask (Hans Rudolph, Kansas, MO) covering the mouth and nose was attached to a bidirectional digital flow valve and fastened by Velcro straps. The gas analyzer and pneumotachograph were calibrated according to the manufacturer's instructions. The ambient temperature ranged from 21 to 23°C and the relative humidity from 50 to 70%.

**$\dot{V}O_2$  at Rest.** The  $\dot{V}O_2$  at rest was assessed on the first visit. After 20 minutes of seated rest in a darkened quiet room, subjects underwent the  $\dot{V}O_2$  assessment in a supine position ( $\dot{V}O_{2000}$ ; Medical Graphics, St. Paul, MN) for 30 minutes, following strict recommendations (6). In brief, this consisted of abstention from physical exercise, alcohol, soft drinks, and caffeine in the 24 hours preceding the assessments, fasting for at least 3 hours, and minimum effort when going to the laboratory. Subjects were laid in a quiet environment (temperature  $22 \pm 1^\circ \text{C}$ , air humidity 50–70%) for 10 minutes, after which the  $\dot{V}O_2$  was measured for 30 minutes. The resting  $\dot{V}O_2$  was defined as the average of the last 5 minutes of steady-state data (6) and used to determine the %  $\text{VO}_2\text{R}$  during AE, besides estimating EPOC and net EE (43).

**Exercise Sessions.** The  $\dot{V}O_2$  was assessed at rest for 10 minutes in a supine position before RE, AE, and CE. Data from the last 5-minute window were used to determine values pre-exercise. The

RE and AE protocols complied with current recommendations for older adults (15). The individual average EE obtained in RE sessions represented the target in AE and CE bouts. After warming up (1 set of 10 reps at 50% of 10 RM in the leg press, seated row, and seated bench press), the individuals performed 3 sets of 10 repetitions at 80% of 10 RM in 8 exercises, interspersed with 2-minute intervals between sets and exercises. The exercises were the leg press, seated row, leg extension, seated bench press, leg curl, seated shoulder press, seated hip adductor, and standing biceps curl. Respiratory data and heart rate were assessed every 10 seconds through telemetry (WIT2410; Cirronet, Dallas, TX). This sampling frequency was necessary for proper visualization of the recovery curve during intervals between sets. However, the respiratory data were transferred every 30 seconds to a spreadsheet specially designed to quantify the total EE in real-time, so that the exercise could be interrupted in AE and CE when the target EE obtained in RE was achieved. In CE, EE targets of each component corresponded to 50% of RE and AE performed in isolation. Immediately after the end of all sessions, subjects were laid on a stretcher and the  $\dot{V}O_2$  postexercise was measured every 10 seconds for 30 minutes.

Based on the  $\dot{V}O_{2\text{peak}}$  obtained in CPET and on  $\dot{V}O_2$  assessed at rest, values corresponding to 60–65% of  $\text{VO}_2\text{R}$  were calculated to determine the intensity of AE bouts (2,15). The absolute  $\dot{V}O_2$  corresponding to % $\text{VO}_2\text{R}$  was used to calculate the associated cycling power by applying the equation:  $\dot{V}O_2 \text{ cycling} = 3.5 + 12.24 \times \text{power} \times (\text{BM})$ , where  $\dot{V}O_2$  is in  $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ , W is the power (watts), and BM is the body mass in kg (2). The AE session began with 5-minute cycling in an electronically-braked cycle ergometer (Inbramed, Porto Alegre, RS, Brazil) with a load corresponding to 50–55% of  $\text{VO}_2\text{R}$  and cadence of  $70 \pm 5 \text{ revs} \cdot \text{min}^{-1}$ . Immediately after this warm-up phase, the individuals exercised with intensity corresponding to 60–65%  $\text{VO}_2\text{R}$  and a cadence of  $70 \pm 5 \text{ revs} \cdot \text{min}^{-1}$ , until reaching the target EE. After AE termination, subjects resumed the supine position, and the  $\dot{V}O_2$  post-exercise was assessed for 30 minutes. The assessment of respiratory data during and after AE followed the same procedures described for RE.



The protocols applied for RE and AE were repeated in the CE session, with RE placed first without intervals between modalities. In each component, individuals exercised until reaching half of the target EE established for the exercise modalities performed in isolation. Furthermore, the pneumotachograph was changed during the transition from the resistance to the aerobic component of CE because of airflow differences during RE and AE.

**Excess Postexercise Oxygen Consumption and Energy Expenditure Calculations.** No difference ( $p > 0.05$ ) and high concordance ( $p < 0.001$ ) were found between  $\dot{V}O_2$  at rest assessed on the first visit ( $0.2 \pm 0.1 \text{ L} \cdot \text{min}^{-1}$ ) vs. pre-exercise assessments (AE:  $0.3 \pm 0.1 \text{ L} \cdot \text{min}^{-1}$ , ICC = 0.92; RE:  $0.3 \pm 0.1 \text{ L} \cdot \text{min}^{-1}$ , ICC = 0.93; CE:  $0.3 \pm 0.1 \text{ L} \cdot \text{min}^{-1}$ , ICC = 0.91). For this reason, we individually calculated the EPOC and net EE as areas under the curves (trapezoidal method) minus pre-exercise  $\dot{V}O_2$  and resting EE, respectively. In both cases, the Weir equation was used (i.e.,  $3.941 \times \dot{V}O_2 + 1.106 \times \dot{V}CO_2$ ) (43). The net  $\dot{V}O_2$  expressed in L was then converted into kcal.

### Statistical Analyses

The sample size for  $F$  tests was calculated a priori using the GPower 3.0.10 (GPower, University of Kiel, Kiel, Germany), considering an effect size = 0.6 for the EPOC,  $1 - \beta = 80\%$ , and  $\alpha = 0.05$ , resulting in a minimum of 10 subjects. The Shapiro-Wilk and Levene tests confirmed the data normality and homoscedasticity, respectively. Therefore, results are presented as mean  $\pm$  SD. The mean values of EE, net EE, and  $\dot{V}O_2$  assessed at rest, pre-exercise, and postexercise (i.e., the EPOC) in AE, RE, and CE were compared through 1-way ANOVAs followed by Tukey post hoc verifications in the event of significant  $F$  ratios. Because the duration of exercise sessions to elicit similar EE was different, it was split into quartiles of time. The  $\dot{V}O_2$  and heart rate data were averaged within each quartile and compared using 2-way ANOVAs with repeated measures, also followed by Tukey post hoc verifications. All calculations were performed using the SPSS software 20.0 (SPSS Inc., Chicago, IL) and statistical significance was fixed at  $p \leq 0.05$ .

### Results

As expected, the total EE during the exercise sessions was similar ( $p > 0.05$ ) in AE ( $126.0 \pm 30.7 \text{ kcal}$ ), RE ( $123.9 \pm 30.6 \text{ kcal}$ ), and CE ( $130.8 \pm 32.6 \text{ kcal}$ ) and felt within the accepted 5% variation range. However, the time to achieve the targeted EE was significantly different across the exercise modalities (RE:  $61.4 \pm 1.9$  minutes  $>$  CE:  $43.3 \pm 5.6$  minutes  $>$  AE:  $26.6 \pm 5.7$  minutes;  $p < 0.001$ ). Table 1 exhibits cardiorespiratory data obtained in the CPET and during the isocaloric exercise bouts. Differences between modalities occurred for all variables ( $p < 0.05$ ). The heart rate (absolute and relative),  $\dot{V}O_2$  (absolute and relative), dioxide carbon production ( $\dot{V}CO_2$ ), and pulmonary ventilation ( $V_E$ ) were greater in AE vs. RE and CE ( $p < 0.01$ ), and in CE vs. RE ( $p < 0.01$ ). However, the RER was higher in CE vs. AE ( $p = 0.022$ ).

Figure 2 depicts the cardiorespiratory and metabolic responses at the end and during the recovery period after the exercise modalities. At the end of the sessions, all outcomes except for RER showed greater values in AE vs. RE and CE. Moreover,  $\dot{V}O_2$ ,  $\dot{V}CO_2$ ,  $V_E$ , and EE were greater in CE vs. RE ( $p < 0.001$ ). However, recovery curves were similar for heart rate,  $\dot{V}O_2$ ,  $\dot{V}CO_2$ , RER, and net EE. Only  $V_E$  differed between AE and RE at

the beginning (5 minutes) of postexercise recovery. Finally, Figure 3 shows that mean EPOC and net EE during recovery were approximately 45% greater in AE ( $p < 0.001$ ) vs. RE or CE.

### Discussion

This study adds to the current knowledge by comparing cardiorespiratory data assessed during and after isocaloric bouts of AE, RE, and CE in older adults. The relative intensity was greater in AE (74%  $\dot{V}O_{2R}$  and 64% heart rate reserve [HRR]) vs. RE (24%  $\dot{V}O_{2R}$  and 18% HRR) and CE (43%  $\dot{V}O_{2R}$  and 34% HRR), whereas the time to achieve the target EE was shorter in AE (27 minutes) than RE (43 minutes) or CE (61 minutes). As for the postexercise recovery, even when matched for the overall EE, AE induced greater EPOC and net-EE ( $8.0 \pm 2.3 \text{ L}$  and  $40.1 \pm 11.7 \text{ kcal}$ , respectively) than RE ( $5.6 \pm 1.2 \text{ L}$  and  $28.1 \pm 5.8 \text{ kcal}$ , respectively) or CE ( $5.4 \pm 2.3 \text{ L}$  and  $26.9 \pm 11.5 \text{ kcal}$ , respectively).

The lowest relative intensity elicited by RE compared with AE or CE was probably because of its intermittent nature. The RE protocol was designed so that each concentric and eccentric phase of movements lasted 1 and 2 seconds per repetition, with sets interspersed with 120-second intervals (2). In consequence, time spent in effort during RE was 12 minutes within the 60-minute sessions. In a few words, the time spent in recovery intervals decreased the mean  $\dot{V}O_2$  and HR during RE. This might have also influenced the relative intensity and EE during CE, which were higher than RE and lower than AE.

As for AE, cycling was performed with intensity corresponding to the lower limit of what is usually classified as vigorous exercise (64% HRR) (16) with a mean duration of 27 minutes. It has been suggested that aerobic gains in older individuals would result from  $150 \text{ min} \cdot \text{wk}^{-1}$  of moderate-intensity exercise (40–59% of HRR or  $\dot{V}O_{2R}$ ) or  $75 \text{ min} \cdot \text{wk}^{-1}$  of vigorous-intensity exercise (60–89% of HRR or  $\dot{V}O_{2R}$ ) (16). In practical terms, this means that performing the present AE protocol  $3 \text{ d} \cdot \text{wk}^{-1}$  would be sufficient for achieving the current recommendations for health-related exercise. This is relevant because older individuals often have difficulties sustaining exercises at high intensities (32). However, prolonged sessions may be demotivating (45). Noteworthy, all subjects completed AE sessions, and there were no complaints of discomfort or excessive fatigue during the study. In addition, AE was more time-efficient for reaching the targeted EE. This was probably because of the greater relative exercise intensity, with the recruitment of larger muscle mass and higher levels of sympathetic activity increasing the postexercise metabolic rate (4). Short strenuous exercise protocols such as HIRT seem to increase the resting  $\dot{V}O_2$  for several hours, in comparison with traditional RE (34).

Studies lack studies comparing the EPOC after different exercise modalities performed with isocaloric conditions, particularly in older adults. We found a single trial investigating the EPOC after RE in older women (13) and 1 trial investigating AE in middle-aged men (25). Farinatti et al. (13) compared the  $\dot{V}O_2$  during and after short RE bouts designed with different exercise orders (large to small muscle groups and vice versa) in young and older women. Significant increases in  $\dot{V}O_2$  occurred along the successive exercises regardless of the exercise order, but the total  $\dot{V}O_2$ , EE, and EPOC, were similar across conditions and age groups, concurring with a previous investigation of the same group including only young women (14). Larsen et al. (25) compared the effects of high-intensity

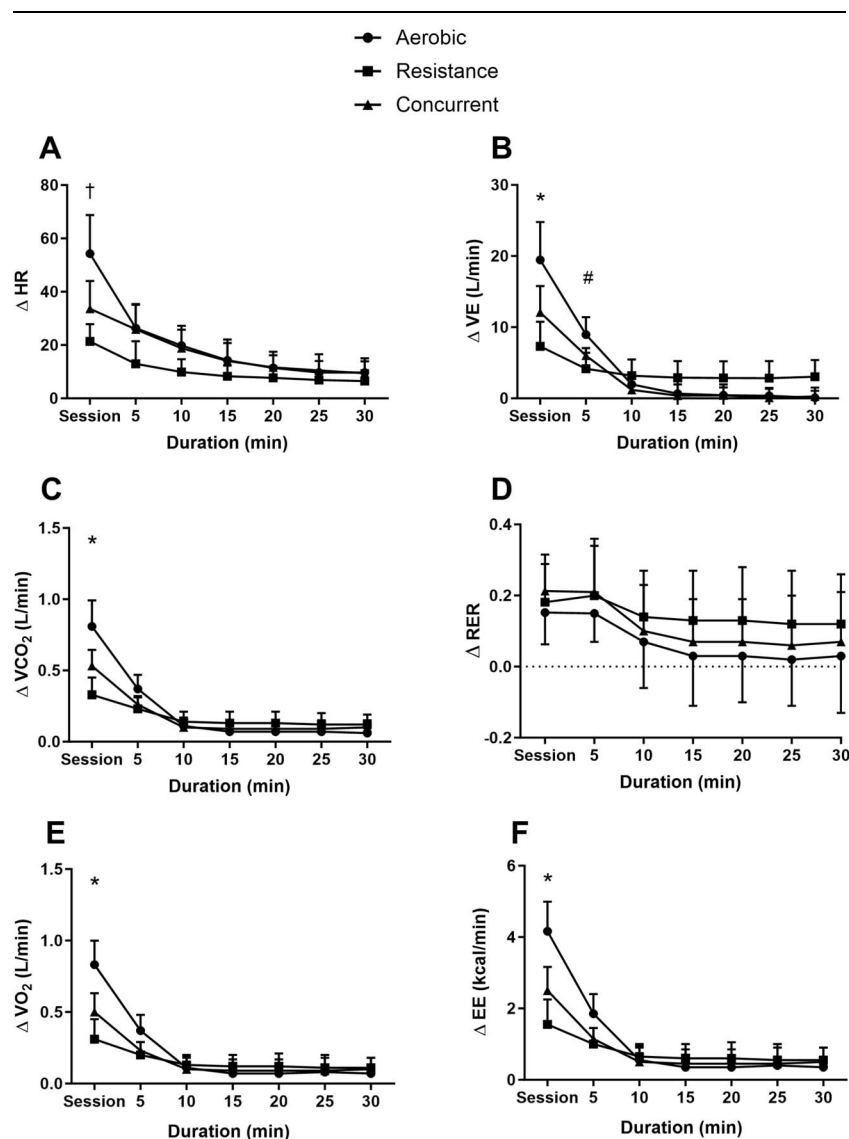
**Table 1****Cardiorespiratory outcomes in the cardiopulmonary exercise testing (CPET) and acute exercise sessions (aerobic, resistance, and concurrent) (mean  $\pm$  SD;  $n = 10$ ).\***

| Variable                                                           | CPET             | Aerobic exercise   | Resistance exercise | Concurrent exercise |
|--------------------------------------------------------------------|------------------|--------------------|---------------------|---------------------|
| HR ( $\text{b}\cdot\text{min}^{-1}$ )                              | 140.3 $\pm$ 22.5 | 114.6 $\pm$ 15.1†‡ | 81.7 $\pm$ 8.1‡     | 93.9 $\pm$ 11.6     |
| HRR (%)                                                            | —                | 64 $\pm$ 6†‡       | 18 $\pm$ 8‡         | 34 $\pm$ 6          |
| $\dot{V}\text{O}_{2\text{max}}$ ( $\text{L}\cdot\text{min}^{-1}$ ) | 1.3 $\pm$ 0.3    | —                  | —                   | —                   |
| $\dot{V}\text{O}_2$ ( $\text{L}\cdot\text{min}^{-1}$ )             | —                | 1.0 $\pm$ 0.2†‡    | 0.5 $\pm$ 0.1‡      | 0.7 $\pm$ 0.1       |
| $\text{VO}_2\text{R}$ (%)                                          | —                | 74 $\pm$ 15†‡      | 24 $\pm$ 9‡         | 43 $\pm$ 13         |
| $\dot{V}\text{CO}_2$ ( $\text{L}\cdot\text{min}^{-1}$ )            | 1.5 $\pm$ 0.4    | 0.9 $\pm$ 0.2†‡    | 0.4 $\pm$ 0.1‡      | 0.7 $\pm$ 0.1       |
| VE ( $\text{L}\cdot\text{min}^{-1}$ )                              | 51.2 $\pm$ 17.2  | 24.2 $\pm$ 5.6†‡   | 12.0 $\pm$ 3.5‡     | 16.6 $\pm$ 3.8      |
| RER                                                                | 1.15 $\pm$ 1.3   | 0.94 $\pm$ 0.07‡   | 0.97 $\pm$ 0.05     | 1.00 $\pm$ 0.1      |

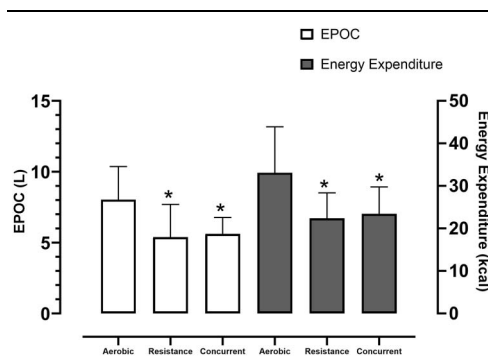
\*HR = heart rate; HRR = heart rate reserve;  $\dot{V}\text{O}_2$  = oxygen consumption;  $\text{VO}_2\text{R}$  = oxygen consumption reserve;  $\dot{V}\text{CO}_2$  = carbon dioxide production; VE = pulmonary ventilation; RER = respiratory exchange ratio; RE = resistance exercise; CE = concurrent exercise.

†Difference vs. RE ( $p < 0.05$ ).

‡Difference vs. CE ( $p < 0.05$ ).



**Figure 2.** Mean  $\pm$  SD values of changes [delta ( $\Delta$ )] in heart rate (HR) (panel A), ventilation (VE) (panel B), carbon dioxide production ( $\dot{V}\text{CO}_2$ ) (panel C), respiratory exchange ratio (RER) (panel D), oxygen uptake ( $\dot{V}\text{O}_2$ ) (panel E), and energy expenditure (EE) (panel F) at the end of isocaloric aerobic, resistance, and concurrent exercise bouts and throughout 30-minute postexercise recovery. \*Significant difference between AE, RE, and CE ( $p < 0.001$ ); †Significant difference between AE and RE ( $p < 0.001$ ); ‡Significant difference between AE vs. RE and CE ( $p < 0.001$ ).



**Figure 3.** Mean  $\pm$  SD values of excess postexercise oxygen consumption (EPOC) and net energy expenditure during 30-minute postexercise recovery after isocaloric aerobic (AE), resistance (RE), and concurrent (CE) exercise. \*Difference between AE vs. RE and CE ( $p < 0.001$ ).

interval training and moderate-intensity continuous AE on postexercise  $\dot{V}O_2$  in men aged 57–11 years with metabolic syndrome. Subjects performed 3 isocaloric AE bouts: 1  $\times$  4-min interval (1-AIT) or 4  $\times$  4-min intervals (4-AIT) at 85–95%  $HR_{max}$  with 3-minute active recovery at 70% of  $HR_{max}$ , and continuous AE at 70%  $HR_{max}$  for 47 minutes (CME). Greater EPOC was found after 4-AIT vs. 1-AIT and CME, indicating that changes in exercise intensity and duration may influence the  $\dot{V}O_2$  during postexercise recovery. The EPOC seems to be also influenced by load intensity in RE—at least 1 trial with young women showed that high-intensity RE would produce similar  $\dot{V}O_2$  but greater EPOC than low-intensity RE with equated work (44). Further investigations are warranted to verify the relative impact of exercise intensity and volume on the EPOC in older vs. younger individuals.

Previous studies comparing the EPOC between exercise modalities in young samples reported greater values in RE than AE (5,17,18). This disagrees with our results indicating that AE produced greater EPOC than RE, which probably resulted from differences in exercise protocols. For instance, the intensity of AE in our experiment was higher than in previous studies. As for RE, previous research often applied strategies that artificially increased the anaerobic component of EE (40) (e.g., pre-exhaustion sets or short recovery intervals)—the need to compensate for the expressive oxygen deficit during RE in those protocols helps to explain the greater EPOC vs. AE. A differential of our study was the application of exercise bouts complying with the guidelines for health-oriented exercise for older subjects (1,15,16,22). This increased the ecological validity of our data, obtained in actual exercise prescription settings.

Comparisons of EPOC between AE, RE, and CE are very scarce. We could find a single trial assessing the  $\dot{V}O_2$  and EPOC in 10 young men ( $26 \pm 0.9$  years) after acute AE, RE, and CE starting with either AE or RE (10). The EPOC magnitude was greater in RE vs. AE and RE, while  $\dot{V}O_2$  remained higher vs. baseline for 10 minutes after AE, 20 minutes after CE (irrespective of the exercise order), and 30 minutes after RE. In that experiment, RE included 3 sets of 7 exercises interspersed with 105-second intervals. Our study applied 120-second rest intervals, which perhaps allowed for additional recovery sufficient to attenuate the postexercise  $\dot{V}O_2$ . As mentioned above, the EPOC after RE may be enhanced when the anaerobic component is not compensated for during the exercise session, particularly when intervals between sets or exercises are too short (11,12,40).

There was no difference in EPOC elicited by RE and CE designed with RE placed before AE. This finding concurs with data from the study by Cutts and Burns (8) showing that CE workouts beginning by either AE or RE produced essentially the same EPOCs in young female participants. However, this is in disagreement with findings from Drummond et al. (10) showing that CE bouts with AE placed before RE increased the EPOC to a greater level than the opposite sequence, despite similar exercise workloads. The authors speculated that placing AE first would allow a better recovery than the inverse sequence, attenuating metabolic and muscular disturbances that contribute to a greater EPOC. The influence of exercise sequence within CE workouts on  $\dot{V}O_2$  along postexercise recovery remains therefore undefined and merits further research.

It is worth noticing that in the study by Cutts and Burns (8), the total EE was greater when AE preceded RE vs. the reverse order, irrespective of the similar EPOC. This is suggestive that AE should be performed before RE when combining the 2 modalities if caloric expenditure is a primary goal, at least in sedentary overweight women. However, performing vigorous AE may compromise the work volume and intensity in subsequent RE, therefore reducing strength and power stimuli during CE (23,27). Because increases in muscle mass, strength, and power are crucial adaptations in older individuals, we placed RE before AE in the present study. Additional research is warranted to ascertain whether changes in CE components influence the EPOC in this particular population.

Strengths of this study included the application of AE, RE, and CE complying with current recommendations of exercise prescription for older adults, which increased the ecological validity of our data in comparison with other studies. Loads were determined based on maximal RE and AE tests and bias because of overall EE on the EPOC was controlled for by applying isocaloric bouts. Moreover, all variables were assessed pre-exercise, ensuring that they were similar in every exercise condition.

However, the sample was relatively small, which limits the generalization of our findings. However, it is worth noting that subjects should not be under medication and have previous experience with exercise protocols. If those restricted criteria helped to avoid bias, they reduced the number of potential eligible volunteers. The RE protocol was designed in a traditional sequence, whereas AE was performed in a cycle ergometer. Other types of exercise, such as circuit RE or treadmill, could produce different results and should be tested in future research. Albeit subjects had previous experience with AE and RE, we must acknowledge that carryover bias might have occurred because of the nonrandomization of RE vs. AE and CE. However, this was necessary for establishing the target EE for AE and CE. Finally, the EE estimation using  $\dot{V}O_2$  and  $CO_2$  data has imprecisions when exercise bouts have a major anaerobic component. Methods using blood lactate have been suggested to be more adequate to assess the contribution of the anaerobic metabolism for the EE (28,29,39,40). Unfortunately, our sample was refractory to blood collections across multiple consecutive sessions, and this precluded the lactate assessment.

In conclusion, the EPOC in older adults was greater after AE vs. RE or CE complying with health-oriented recommendations for this population and matched for caloric expenditure. Moreover, in AE the targeted EE was achieved in shorter bouts of higher relative intensity, followed by CE and RE. This suggests that the exercise intensity would be more determinant of the  $\dot{V}O_2$  postexercise than the overall volume reflected by EE. Further research is warranted to clarify if these data replicate in other age groups and exercise modalities performed with different intensities and volumes.

## Practical Applications

The greater EPOC after AE indicates that at least in older adults this modality may be preferred over RE or CE when the purpose is to increase the EE in a short time after training sessions. Moreover, moderate-intensity continuous AE proved to be more time-efficient than RE and CE to achieve a target EE. In practical terms, AE could remain the priority choice vs. RE or CE in the short run within exercise routines designed as adjuvants of metabolic control or weight management interventions. However, it is feasible that RE should be chosen in the long term because of gains in muscle mass, which possibly increases EE at rest and during exercise.

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